Data Flow Diagram

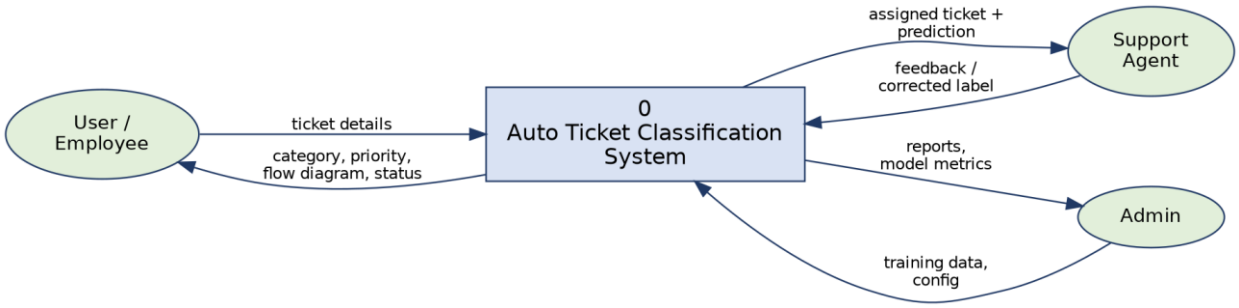
Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	2 Marks

The Data Flow Diagram (DFD) shows how ticket data moves between external entities, processes and data stores.

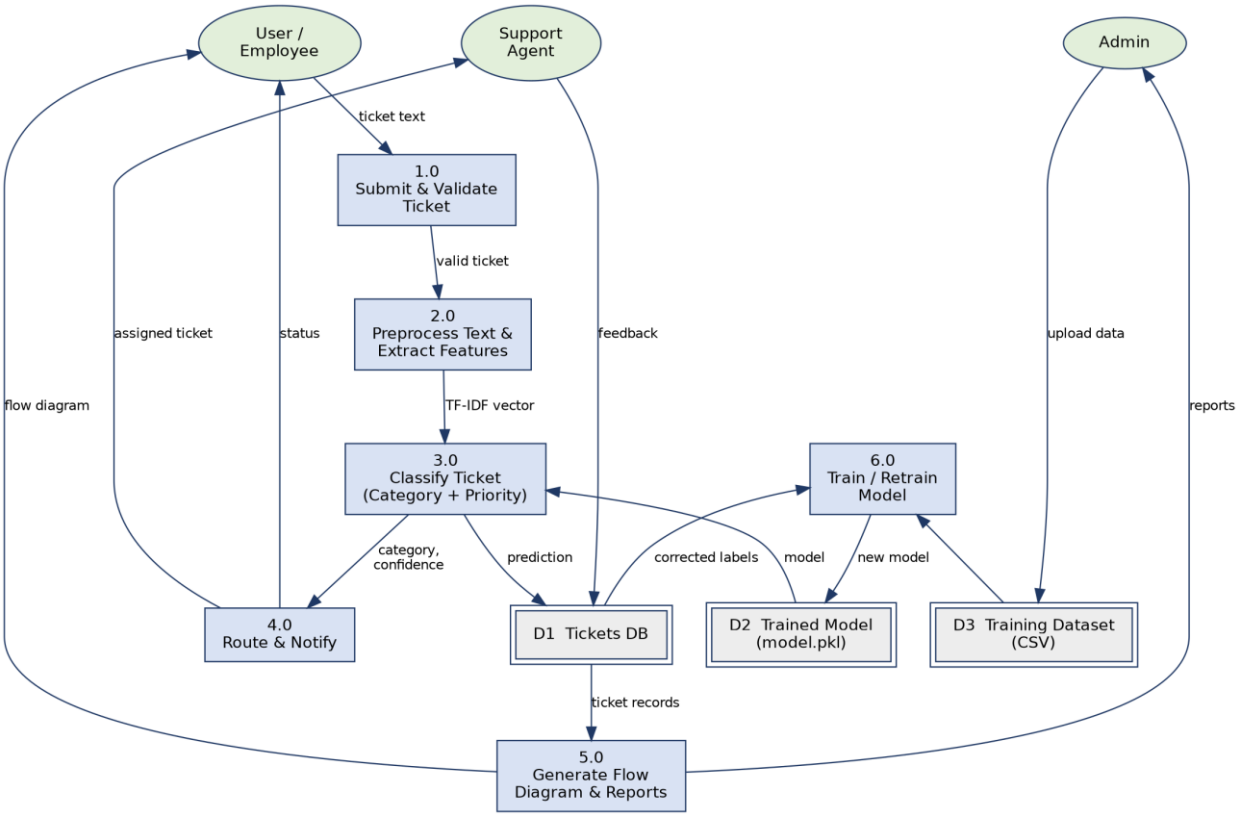
DFD Symbol Legend

Symbol	Name	Description
Oval	External Entity	A person or system outside the project that sends or receives data (User, Support Agent, Admin).
Rectangle with numbered header (1.0, 2.0)	Process	An activity that transforms incoming data into outgoing data (Classify Ticket).
Double-line rectangle (D1, D2)	Data Store	A place where data is held for later use (Tickets DB, Trained Model, Dataset).
Labelled arrow	Data Flow	The movement of data, labelled with what is passed.

Level 0: Context Diagram



Level 1 DFD



Process description

ID	Process	Description
1.0	Submit & Validate Ticket	Receives title and description from the user and checks that the text is not empty and within the length limit.
2.0	Preprocess Text & Extract Features	Cleans the text (lowercase, punctuation, stop-words, lemmatisation) and converts it to a TF-IDF vector.
3.0	Classify Ticket	Loads the trained model (D2), predicts category and confidence, and assigns priority.
4.0	Route & Notify	Sends the ticket to the responsible team, updates status and notifies user and agent.
5.0	Generate Flow Diagram & Reports	Builds the ticket flow diagram and summary reports from stored records.
6.0	Train / Retrain Model	Trains the model on the dataset (D3) and on corrected labels from feedback (D1).